3.5 Carbonyl Compounds

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.5 Carbonyl Compounds
Difficulty	Easy

Time allowed: 30

Score: /19

Percentage: /100

Question 1a

State the reactants that can be oxidised to give an aldehyde and a ketone as products.

[2 marks]

Question 1b

State the oxidation products, where appropriate, of an aldehyde and a ketone.

[2 marks]

Question 1c

Describe how you could use 2,4–DNPH and melting point data to determine the identity of an unknown compound.

[3 marks]

Question 2a

Butanone can react with HCN.

Complete the reaction mechanism shown in Fig. 2.1 by drawing four curly arrows.

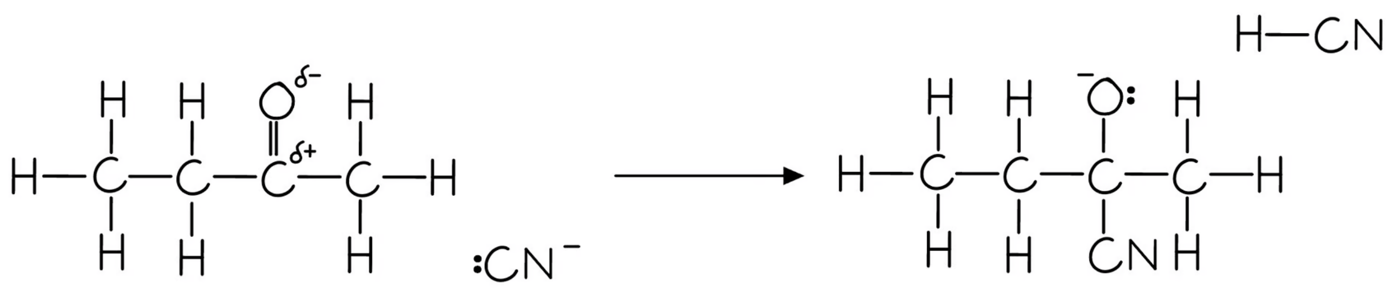


Fig. 2.1

[4 marks]

Question 2b

Name the organic produced in part **(a)**.

[1 mark]

Question 2c

Explain why butanone has no positional isomers.

[1 mark]

Question 3a

Table 3.1 shows the initial observations for the reaction for Tollens' reagent with aldehydes and ketones.

Table 3.1

Ketone		Aldehyde	
Initial observation	**Final observation**	**Initial observation**	**Final observation**
Colourless solution		Colourless solution	

Complete the final observations for aldehydes and ketones.

[2 marks]

Question 3b

Table 3.2 shows the initial observations for the reaction for Fehling's with aldehydes and ketones.

Table 3.2

Ketone		Aldehyde	
Initial observation	**Final observation**	**Initial observation**	**Final observation**
Blue solution		Blue solution	

Complete the final observations for aldehydes and ketones.

[2 marks]

Question 3c

Ethanal and propanal are reacted separately heated with an alkaline solution of iodine.

i)
State which aldehyde will give a yellow precipitate for this reaction.

[1]

ii)
Explain your answer to part **(c)(i)**.

[1]

[2 marks]

Model Answers: Easy

1a

a) The reactants that can be oxidised to give an aldehyde and a ketone as products are:

- A primary alcohol to produce an aldehyde; [1 mark]
- A secondary alcohol to produce a ketone; [1 mark]

[Total: 2 marks]

- A primary alcohol has the hydroxyl group at the end of a hydrocarbon chain which oxidises to form an aldehyde
- A secondary alcohol has the hydroxyl group in the middle of a hydrocarbon chain which oxidises to form a ketone

1b

b) The oxidation products, where appropriate, of an aldehyde and a ketone are:

- An aldehyde oxidises to a carboxylic acid; [1 mark]
- A ketone does not undergo further oxidation (without destruction of the ketone structure); [1 mark]

[Total: 2 marks]

- Primary alcohols oxidise to form aldehydes
- Aldehydes oxidise to form carboxylic acids
    - Primary alcohol + [O] → aldehyde + [O] → carboxylic acid
      or
      Primary alcohol + 2[O] → carboxylic acid
- Secondary alcohols oxidise to ketones
- Ketones do not further oxidise
    - Secondary alcohol + [O] → ketone

1c

c) You could use 2,4-DNPH and melting point data to determine the identity of an unknown compound as follows:

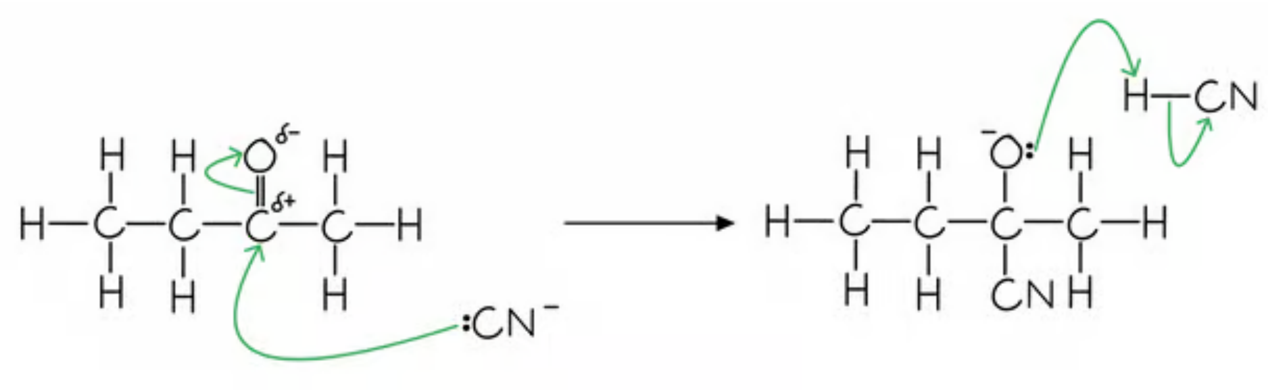
- Carbonyl compounds form a crystalline product upon reacting with 2,4-DNPH
  OR
  Carbonyl compounds form a (deep) orange ppt; [1 mark]
- Precipitate purified
  OR
  Recrystallised; [1 mark]
- The melting point of this crystalline product can be measured
  AND
  This value can be compared to known melting points of crystalline products of different carbonyl compounds; [1 mark]

[Total: 3 marks]

- Brady's reagent is the common name for 2,4-DNPH dissolved in methanol and sulfuric acid
- The crystalline products are sometimes called 2,4-DNPH derivatives
- A derivative of a compound is a similar compound to the original or one that has been made from it

2a

a) The completed mechanism is:



- Curly arrow from lone pair on CN to δ+ carbon atom; [1 mark]
- Curly arrow from C=O bond to δ- O atom; [1 mark]
- Curly arrow from lone pair on O atom to H atom in HCN molecule; [1 mark]
- Curly arrow from H-CN bond to C atom; [1 mark]

[Total: 4 marks]

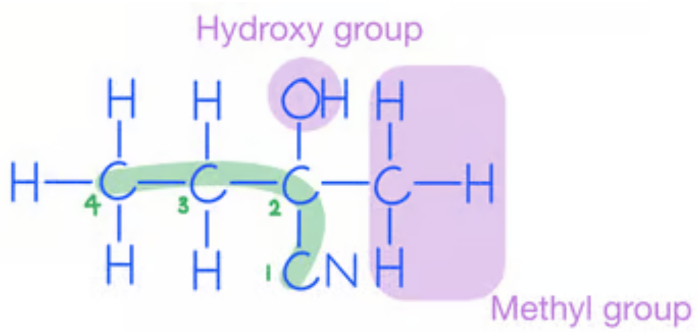
- You are asked to draw 4 arrows in order to complete this mechanism
    - There is one mark for each arrow
- The nucleophilic addition of hydrogen cyanide to carbonyl compounds is a two-step process
- In step 1 the cyanide ion attacks the carbonyl carbon to form a negatively charged intermediate
- In step 2 the negatively charged oxygen atom in the reactive intermediate quickly reacts with aqueous H^+

2b

b) The product is:

- 2–hydroxy-2–methylbutanenitrile; [1 mark]

[Total: 1 mark]



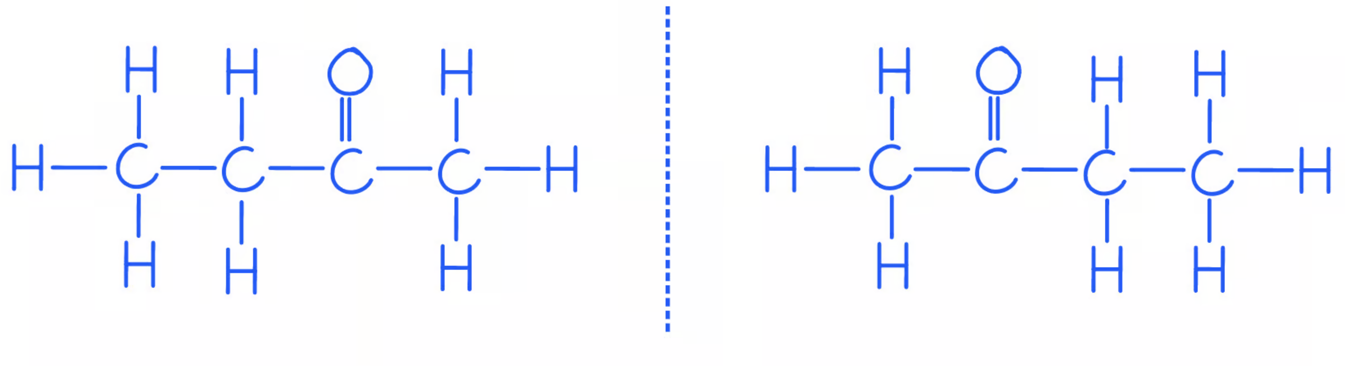
2c

c) Butanone has no positional isomers because:

- The ketone group can only be attached to carbon 2 / the second carbon; [1 mark]

[Total: 1 mark]

- **Remember:** The functional group of a ketone is a carbonyl / C=O bond in the middle of a hydrocarbon chain
- For butanone, the ketone group can only go on carbon 2 / the second carbon
    - If you draw butan-2–one and butan-3–one, you will see that they are mirror images of one another



- This is why there is no butan-2-one or butan-3-one as it is not necessary to include the number that places the carbonyl group

3a

a) The complete final observations for aldehydes and ketones with Tollens' reagent are:

Ketone		Aldehyde	
Initial observation	**Final observation**	**Initial observation**	**Final observation**
Colourless solution	**Colourless solution**; [1 mark]	Colourless solution	**Silver mirror**; [1 mark]

[Total: 2 marks]

- You are expected to know the positive tests for an aldehyde with Tollens' reagent
- It is helpful to also know the original observations for Tollens' reagent
  - Tollens' reagent is a colourless solution

3b

b) The observations are:

Ketone		Aldehyde	
Initial observation	**Final observation**	**Initial observation**	**Final observation**
Blue solution	**Blue solution**; [1 mark]	Blue solution	**Red precipitate**; [1 mark]

[Total: 2 marks]

- When warmed with an aldehyde, the aldehyde is oxidised to a carboxylic acid and the Cu^{2+} ions from Fehling's solution are reduced to Cu^{+} ions
- The clear blue solution turns opaque due to the formation of a red precipitate, copper(I) oxide
- Ketones cannot be oxidised and therefore give a negative test when warmed with Fehling’s solution and the solution remains blue

3c

c)

i) The aldehyde that will give a yellow precipitate is:

- Ethanal; [1 mark]

ii) The reason why is:

- Contains a methyl ketone group; [1 mark]

[Total: 2 marks]

- Methyl ketones are compounds that have a CH_3CO-group
- Ethanal also contains a CH_3CO- group and therefore also forms a yellow precipitate with iodoform
  - Ethanal is the only aldehyde that gives a positive result (yellow precipitate) when it undergoes the iodoform reaction
    - The yellow precipitate is CHI_3
- Propanal does not contain a methyl ketone group so will not give a yellow precipitate